

29.09.2024

Practical - 11 (a)

STATIC ROUTING

AIM:-

To simulate and configure static routing with primary and backup routes using Cisco Packet Tracer and verify connectivity between different networks.

PROCEDURE:-

- 1) Open Cisco Packet Tracer and create the given topology with Router 0, Router 1, Router 2, switches and PCs.
- 2) Configure the IP addresses on all the routers interfaces and PC's according to the given topology.
- 3) Open the CLI of Router 0 and enter configuration mode.
- 4) Configure static routes for networks 30.0.0.0/8 and 50.0.0.0/8 using Router 1 as the primary route and Router 2 as the backup route.
- 5) Configure a separate host route for 30.0.0.100/32 with Router 2 as the primary route and Router 1 as the backup route.
- 6) On Router 1, configure static routes for networks 10.0.0.0/8 and 40.0.0.0/8, with primary and backup routes.
- 7) On Router 2, configure static routes for networks 10.0.0.0/8 and 30.0.0.0/8.
- 8) Use the command: `show ip route static` to verify the configured static routes.
- 9) Use the `tracert` command from a PC to verify the path taken by packets to the destination network.
- 10) Disconnect the primary link between R0 and R1 and use `tracert` again to verify that the backup route becomes active.
- 11) Static routes can be removed using: `no ip route`.
- 12) Verify the routing table again using `show ip route static`.

```

PC0
Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.0.0.100

Pinging 30.0.0.100 with 32 bytes of data:

Request timed out.
Reply from 30.0.0.100: bytes=32 time=2ms TTL=126
Reply from 30.0.0.100: bytes=32 time=18ms TTL=126
Reply from 30.0.0.100: bytes=32 time=24ms TTL=126

Ping statistics for 30.0.0.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 24ms, Average = 14ms

C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.0.0.1
  1  1 ms    0 ms    0 ms    40.0.0.2
  2  3 ms    2 ms    2 ms    20.0.0.2
  3  2 ms    5 ms    0 ms    30.0.0.100

Trace complete.

C:\>

```

```

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/1/1
Router(config-if)#shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down

```

```

C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.0.0.1
  1  3 ms    1 ms    1 ms    20.0.0.2
  2  0 ms    0 ms    1 ms    30.0.0.100

Trace complete.

```

```
Router(config-if)#interface serial 0/1/1
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

```
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
```

```
C:\>tracert 30.0.0.100
```

```
Tracing route to 30.0.0.100 over a maximum of 30 hops:
```

1	0 ms	0 ms	0 ms	10.0.0.1
2	1 ms	7 ms	4 ms	40.0.0.2
3	9 ms	7 ms	1 ms	20.0.0.2
4	2 ms	0 ms	4 ms	30.0.0.100

```
Trace complete.
```

```
C:\>
```

RESULT:-

Thus, the routing with primary and backup routes was successfully configured and simulated using the Cisco Packet Tracer.

29.09.2026

PRACTICAL - 11(b)

RIP (Routing Information Protocol)

AIM:-

To simulate and configure RIP (Routing Information Protocol) using Cisco Packet Tracer and verify dynamic routing between different networks.

PROCEDURE:-

- 1) Open Cisco Packet Tracer and create the given topology with Router0, Router1, Router2, PC0 and PC1.
- 2) Assign IP addresses to PCs and router interfaces according to table.
- 3) Configure the FastEthernet and Serial interfaces of all routers using the assigned IP addresses.
- 4) Use no shutdown to activate all router interfaces.
- 5) For DCE serial interfaces, configure the required clock rate and bandwidth.
- 6) Configure RIP on R0 using:-

```
router rip
network 10.0.0.0
network 192.168.1.252.
```
- 7) Configure RIP on R1 using:-

```
router rip
network 192.168.1.244
network 192.168.1.248
```
- 8) Configure RIP on R2 using:-

```
router rip
network 20.0.0.0
network 192.168.1.252.
network 192.168.1.244
```
- 9) Allow RIP to exchange routing information automatically between the routers.
- 10) From PC0, use the ping command to verify connectivity with PC1.
- 11) Use the tracer command to observe the path followed by packets between the networks.
- 12) Disconnect one of the available routes and use tracer again to verify that RIP automatically selects the alternate one.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=19ms TTL=126
Reply from 10.0.0.2: bytes=32 time=19ms TTL=126
Reply from 10.0.0.2: bytes=32 time=18ms TTL=126
Reply from 10.0.0.2: bytes=32 time=19ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 19ms, Average = 18ms

C:\>tracert 10.0.0.2

Tracing route to 10.0.0.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    20.0.0.1
  1  11 ms   13 ms   12 ms   192.168.1.254
  2   9 ms    5 ms    5 ms   10.0.0.2

Trace complete.

C:\>
```

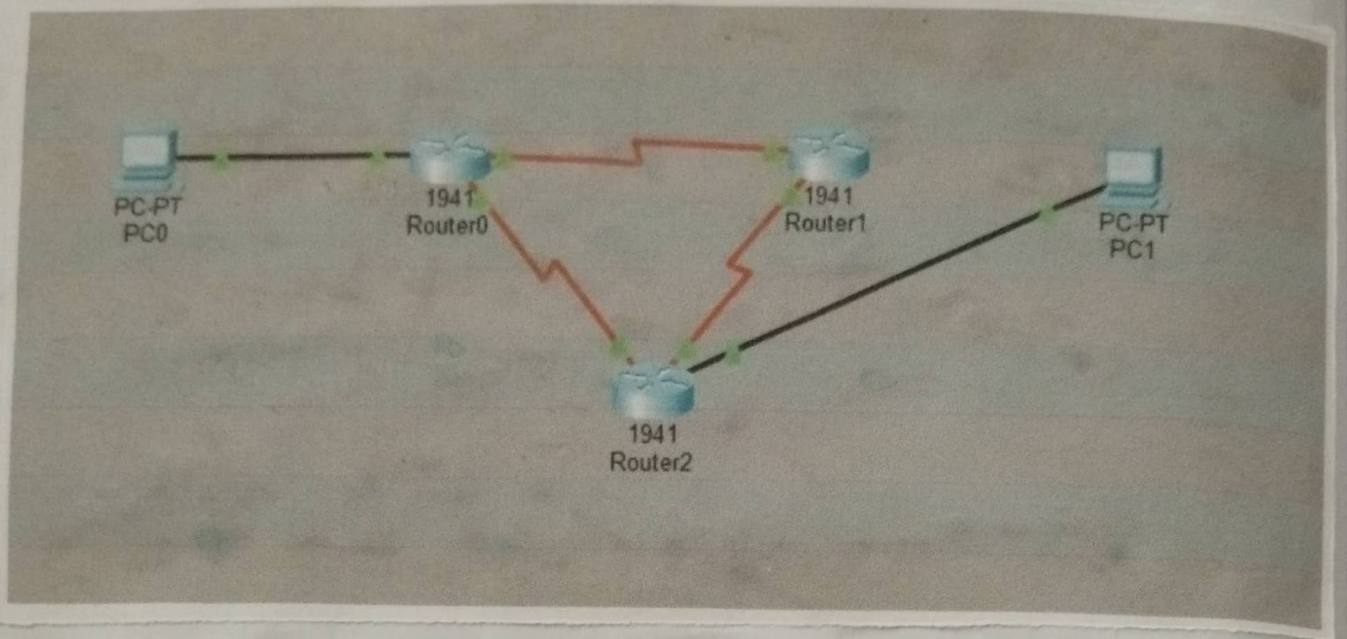
```
C:\> tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.0.0.1
  1   6 ms    8 ms    7 ms   192.168.1.253
  2   5 ms   10 ms    7 ms   20.0.0.2

Trace complete.
```

ping 10.0.0.2 has brought up the connection and the connection is now established



✓
OK

Result:-

Thus RIP was successfully configured and simulated using Cisco Packet Tracer.